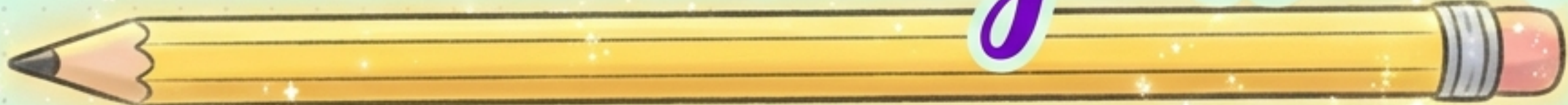




Decoding Arithmetic Progressions



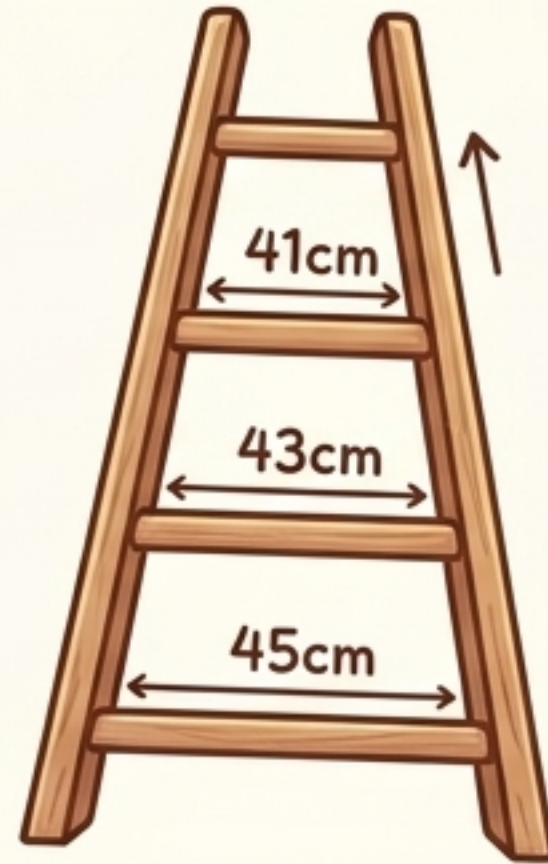
THE ULTIMATE AESTHETIC CHEAT SHEET
FOR MATH PATTERNS.



Swipe to unlock
the secret code
of numbers! →



Nature's blueprints follow strict numerical patterns.



Rungs on a ladder decreasing evenly by 2cm: 45cm, 43cm, 41cm...

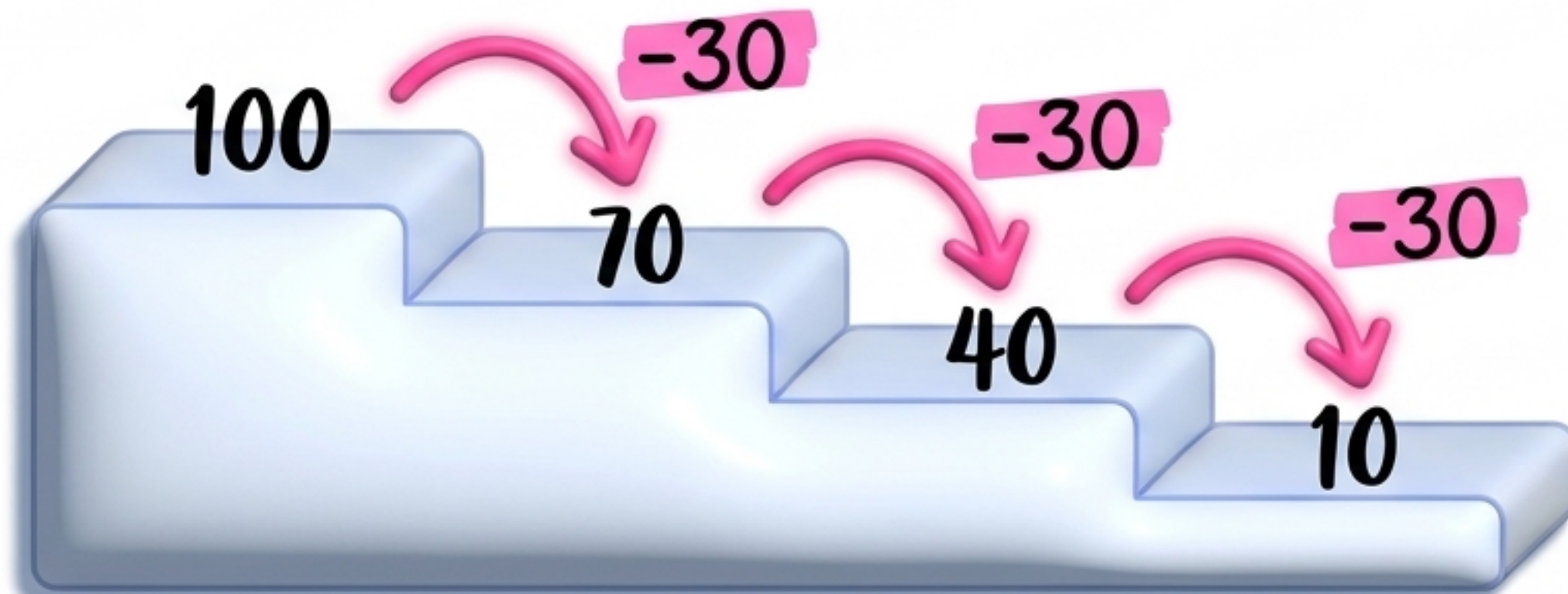


Shakila's money box: ₹100, ₹150, ₹200... saving ₹50 more every year!

Math isn't just random numbers; it's the predictable rhythm of the world around us.

What is an A.P.?

An Arithmetic Progression is a list of numbers where each term is created by adding a **fixed number** to the term right before it (except for the first term).



The Anatomy of an A.P.



a (or a_1)

= The First Term
(The starting line!)

d

= The Common
Difference
(The jump size!)

a_n

= The n th Term
(The number sitting at
a specific position).

n

= The Position/
Term
Number
(e.g., $n=4$).



The Golden Rule of d

d is always found by subtracting a term from the one right after it:

$$d = a_{k+1} - a_k$$



Positive (+)

Sequence goes UP.

1, 2, 3, 4... ($d = 1$)



Negative (-)

Sequence goes DOWN.

100, 70, 40, 10... ($d = -30$)



Zero (0)

Sequence stays FLAT.

3, 3, 3, 3... ($d = 0$)





Two Flavors of A.P.



Finite A.P.



- Has a final destination.
- There is a specific last term (denoted as l or a_n).
- Example: Heights of students in a line (147cm, 148cm... 157cm).

Infinite A.P.



- Goes on forever.
- No last term (ends with the famous "...").
- Example: Odd integers (1, 3, 5, 7, 9...).



The Time Travel Formula

Want to find the 100th term without counting step-by-step? Fast-forward instantly!


$$a_n = a + (n - 1)d$$


- a_n : The destination number.

- a : Where you started.

- $(n-1)$: The number of 'jumps' you take.

- d : The size of each jump.

Pro-Tip: To find the 25th term, just do $a + 24d$. Always one jump less than the position!

The Ultimate Accumulator (S_n)

How to add up massive lists of numbers in seconds.

The Standard Code



Use when you know:
 a , d , and n

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

Perfect for when you know the jump size but don't know the final destination.



The Shortcut Code

Use when you know:
First & Last Terms

$$S_n = \frac{n}{2} (a + l)$$

If they give you the last term (l or a_n), use this instant shortcut!

Secret Studygram Hacks 🤫

HACK

The Sum of First n
Positive Integers!

$$1 + 2 + 3 + \dots + n$$

Shortcut formula:

$$S_n = \frac{n(n+1)}{2}$$

The Missing Term Trick

If you have the sums,
but need the term:

$$a_n = S_n - S_{n-1}$$

(The nth term is the sum
of n terms minus the sum
of the previous terms!)

Arithmetic Mean

If numbers a, b, c
are in an A.P., the
middle guy is just
the average!

$$b = \frac{(a+c)}{2}$$

The Core Rule

- $d = a_2 - a_1 = a_3 - a_2$
- d can be $+$, $-$, or 0 .

The nth Term

- $a_n = a + (n-1)d$
- Fast-forwards to any position!

Arithmetic Progressions

Sum Formula (Standard)

- $S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$

Sum Formula (Shortcut)

- $S_n = \left(\frac{n}{2}\right)(a + l)$
- Use when the last term is known!